



Predicting Used-Car Prices

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github.com/MurphyChen0413/data1030final_project





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Predicting Used-Car Prices in the UK Market

Goal

- Develop a data-driven pricing model for the UK market

Motivation

- Helps buyers find fair prices
- Helps dealers optimize inventory & pricing

ML Task

- **Supervised Learning** - Regression
- Target variable: **price**

Dataset Challenges

- Large database (~550K rows)
- Missing Values



Data Overview

Dataset Source

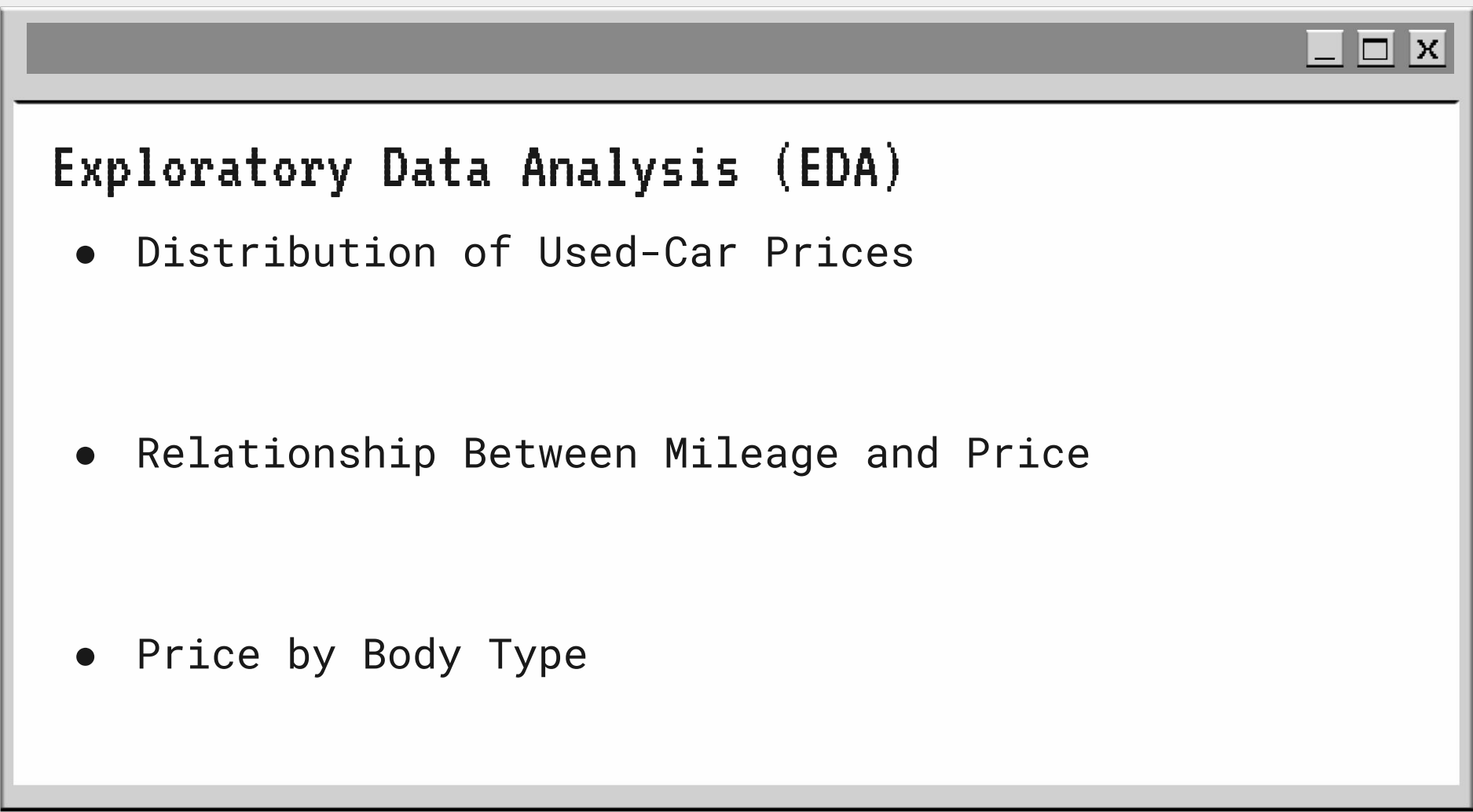
- DVM-CAR Dataset
- Created by researchers from the University of Glasgow & USC
- Collected through used-car advertisements from UK online platforms

Dataset Size

- ~536,000 rows * 25 columns
- Covers 900 car models from 100 automakers
- Spans from 2000 - 2020 registration years

Key Features

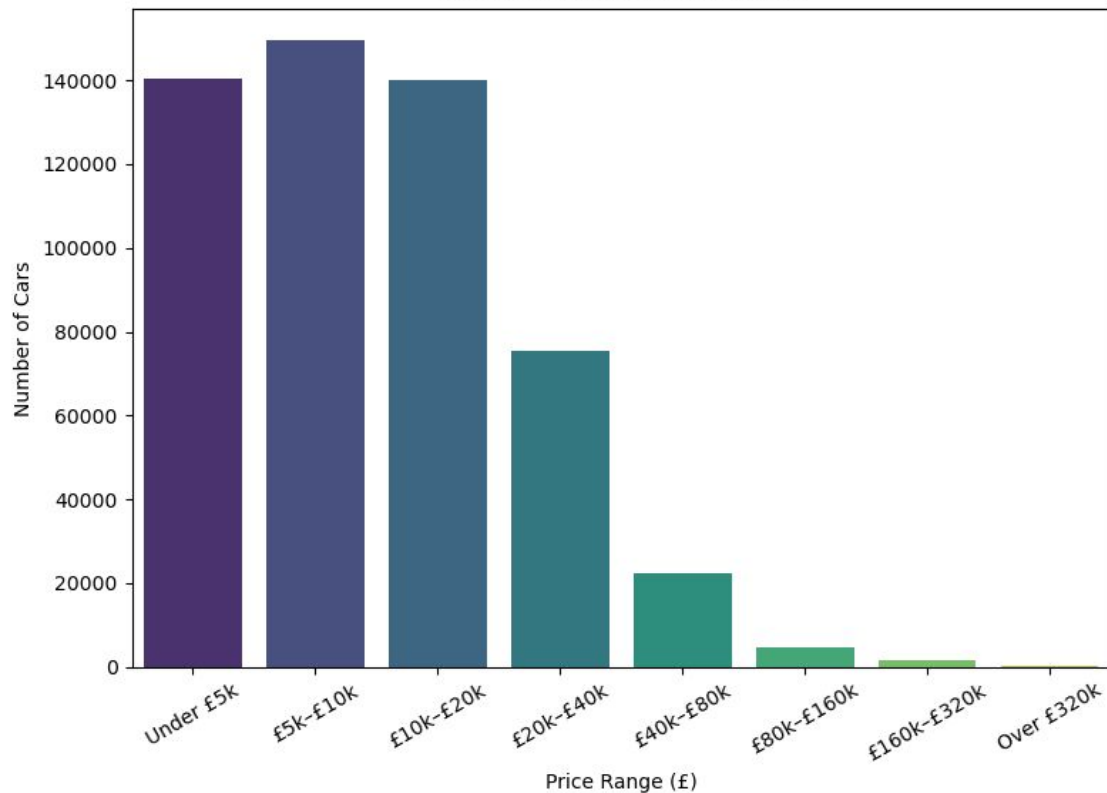
- Numeric: Price, Mileage, Engine Size, Registration Year
- Categorical: Brand, Model, Fuel Type, Gearbox, Body Type, Color

A window with a title bar containing three icons (minimize, maximize, close) on the right. The main content area has a title and a bulleted list.

Exploratory Data Analysis (EDA)

- Distribution of Used-Car Prices
- Relationship Between Mileage and Price
- Price by Body Type

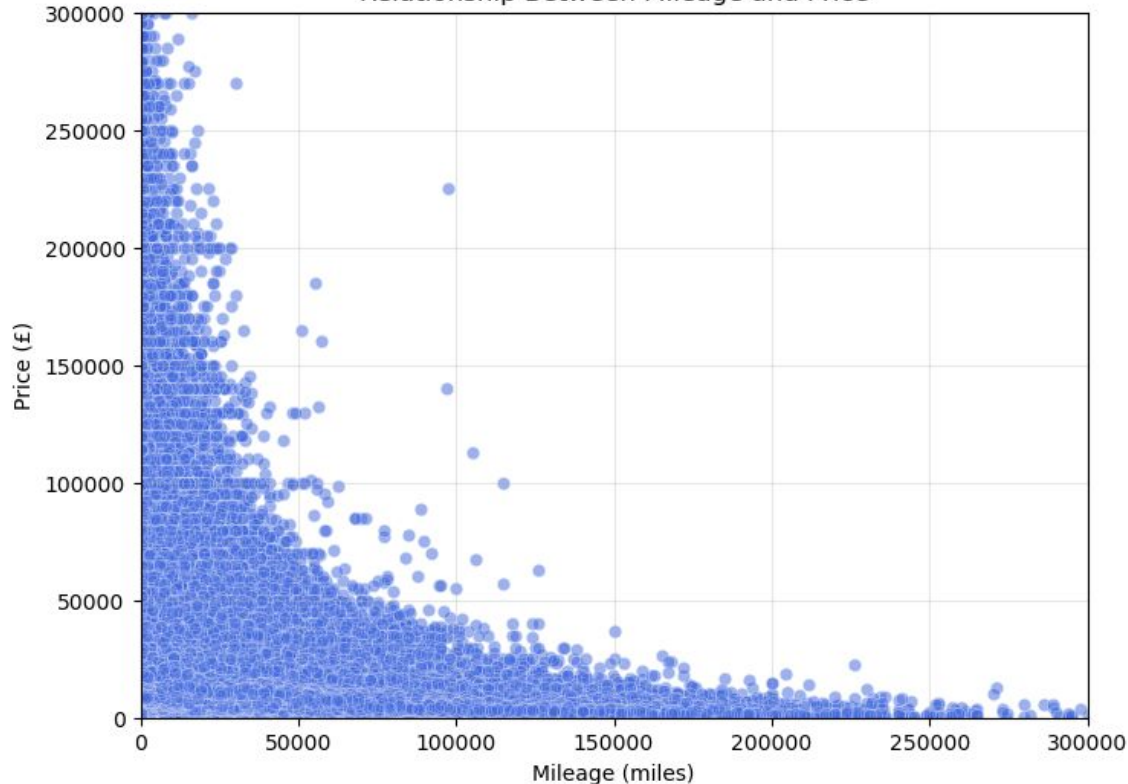
Distribution of Used-Car Prices



Right-skewed:

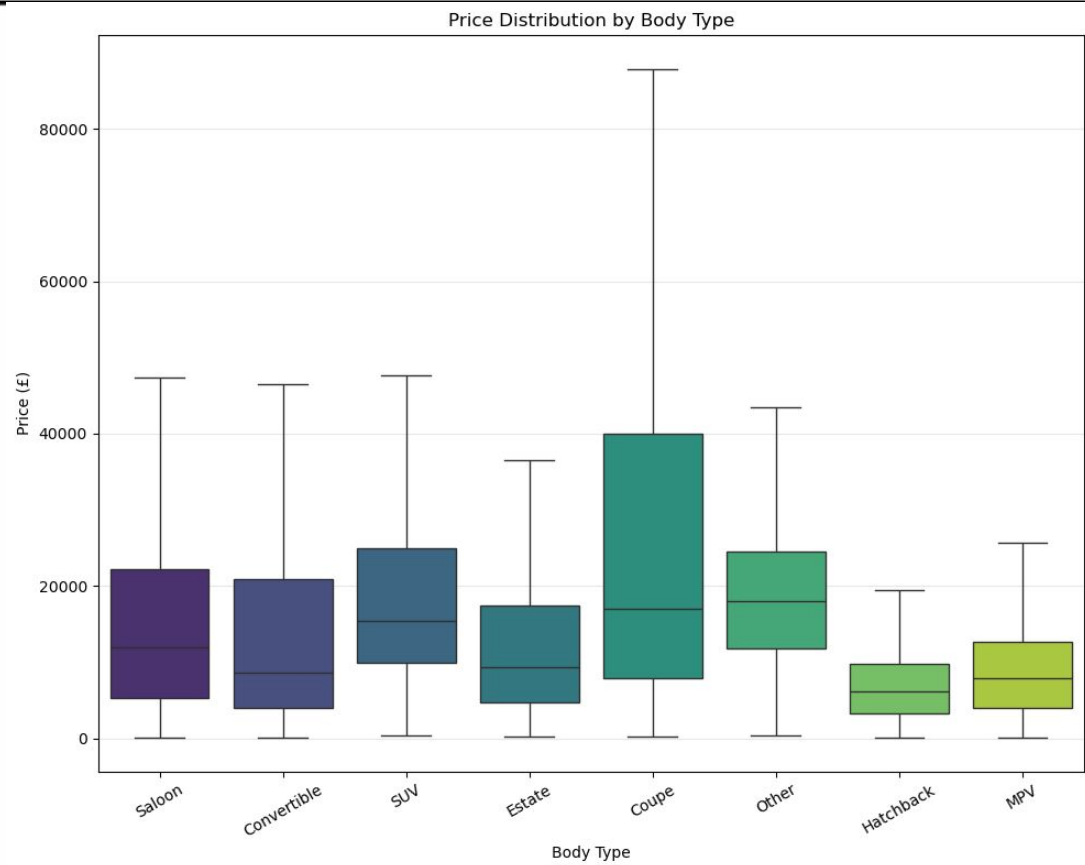
- most listings under £20k
- £5k-£10k range dominates
- Only a few luxury outliers above £80k
- Indicates high variability of used-car prices

Relationship Between Mileage and Price



Mileage vs Price

- Strong negative correlation
- Price drop sharply below 50k miles
- Nonlinear depreciation
- High variance at low mileage



Price by Body Type

- Coupes & SUVs: highest average prices
- Hatchbacks & MPVs: lowest, narrow price range
- Large variation: reflects lifestyle-driven price differentiation

Data Preprocessing

Data Cleaning

- Removed redundant columns and extra spaces in headers
- Converted mixed data types (e.g., price, engine_size) to numeric
- Cleaned categorical columns (e.g., fuel_type & color)

Missing Values

- Did not drop rows (536K rows)
- Imputed numeric columns with median
- Imputed categorical columns with most frequent ones

	missing_count	missing_fraction
color	43750	8.15
seat_num	12948	2.41
door_num	9106	1.70
engine_size	4128	0.77
price	2290	0.43
mileage	2110	0.39
bodytype	1908	0.36
fuel_type	818	0.15
gearbox	334	0.06
reg_year	14	0.00



Data Splitting

- Train/Test Split: 80% training, 20% testing
 - Ensure balanced evaluation and prevents overfitting
- Stratified by Price range: preserves market distribution
- Encoding: One-hot encoding for categorical variables
- Scaling: apply StandardScaler to standardize numeric variables



Thanks :)